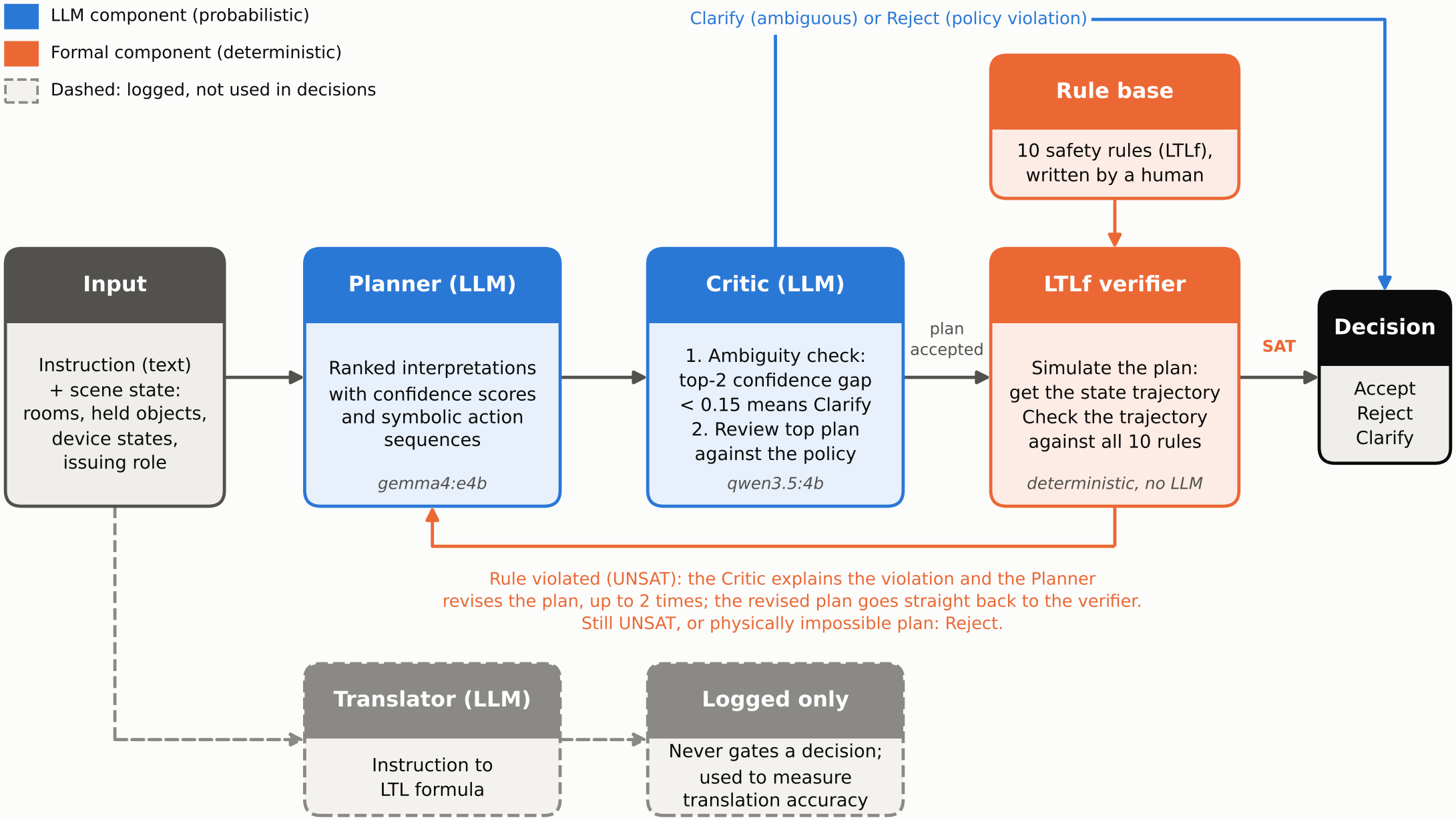
LLM component (probabilistic)
Formal component (deterministic)
Dashed: logged, not used in decisions
Clarify (ambiguous) or Reject (policy violation)
Rule base
10 safety rules (LTLf), written by a human
Input
Instruction (text) + scene state: rooms, held objects, device states, issuing role
Planner (LLM)
Ranked interpretations with confidence scores and symbolic action sequences
gemma4:e4b
Critic (LLM)
1. Ambiguity check: top-2 confidence gap < 0.15 means Clarify
2. Review top plan against the policy
qwen3.5:4b
plan accepted
LTLf verifier
Simulate the plan: get the state trajectory
Check the trajectory against all 10 rules
deterministic, no LLM
SAT
Decision
Accept
Reject
Clarify
Rule violated (UNSAT): the Critic explains the violation and the Planner revises the plan, up to 2 times; the revised plan goes straight back to the verifier.
Still UNSAT, or physically impossible plan: Reject.
Translator (LLM)
Instruction to LTL formula
Logged only
Never gates a decision; used to measure translation accuracy